
AUTOMATIC VISUAL CLASSIFICATION AND INDIVIDUAL IDENTIFICATION OF EARTHWORMS

DEEP LEARNING FOR VISUAL RECOGNITION

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ABSTRACT

We conducted a study utilizing three distinct neural networks—ResNet-50, ConvNeXT, and Swin Transformer—to classify a dataset of 1,478 images of earthworms into five different genera. Additionally, we explored the influence of image background area on classification performance. Employing the Swin Transformer on square images with minimal background area, we achieved a classification accuracy of 75.9% and an F1-score of 0.7 on the test set. Furthermore, we implemented ResNet-50 on a dataset comprising 4,268 images corresponding to 775 individual earthworms, with the objective of determining whether different images represented the same earthworm. This analysis was based on the "triplet loss" function, resulting in an accuracy of approximately 90% on both the training and test sets. A visual examination of the embedding points, following t-SNE reduction, demonstrated that these points were closely grouped together, confirming the model's effectiveness.

Keywords Deep Learning · Earthworm classification · Individual earthworm identification

1 Introduction

In biology, organisms sharing similar characteristics are systematically grouped at different hierarchical levels known as taxonomic ranks. Among these, there is the genus level. Identifying the genus of earthworms is especially important for assessing soil health and for efforts to restore soil that has been depleted of minerals and nutrients due to pollution or agricultural use (Schon et al. [2021], Al-Maliki et al. [2021]). Traditionally, genus classification of earthworms requires experts to euthanize the specimen and conduct microscopic analysis (Sherlock [2012]). This process is time-consuming, resource-intensive, and necessitates killing the organism. Therefore, developing an automatic method for classifying earthworms based on images could conserve time, resources, and the lives of the organisms. This project aims to use deep learning to automatically classify earthworm images into five genera.

Additionally, behavioural studies of soil organisms often involve recording their activity with cameras, enabling subsequent video or image analysis (Banerjee et al. [2023], Djerdj et al. [2020]). However, the field of view is a limitations of cameras, which means that earthworms may leave and re-enter the frame, making it difficult to determine whether the same individual is being observed. Researchers sometimes assume that a brief absence from the frame indicates the return of the same individual, although this assumption is not always accurate. An automatic method for identifying individual earthworms in such scenarios would be beneficial. Additionally, the objective of this project is to automatically identify images of the same earthworm specimen as belonging to the same earthworm in different images of the same earthworm.

Furthermore, earthworm classification presents additional challenges due to their highly flexible and dynamic body shapes. Earthworms can occupy a substantial proportion of image pixels or, alternatively, leave much of the background visible, complicating image-based analysis. Studies have examined how background removal can

influence classification performance by eliminating extraneous information (Liang et al. [2023], Xi et al. [2019]). However, to our knowledge, no study has explored the effect of the ratio of non-background to background pixels on classification performance for flexible or branching objects, such as earthworms or neurons. This project aims to change the proportions of background areas in earthworm images to investigate the influence of background area over the classification performance.

Consequently, this study will evaluate the classification capabilities of three deep learning models —Resnet 50 (Chen et al. [2017]), ConvNeXt (Liu et al. [2022]), and Swin Transformer (Liu et al. [2021])— in the classification of earthworm images in five genus classes, *Eisenia*, *Eiseniella*, *Aporrectodea*, *Alolobophora*, *Lumbricus*, based solely on earthworm information (i.e., without background information). We will refer to these experiments as "Classification Experiments". Furthermore, we will investigate the impact of background area on the models' classification performance. We will refer to these experiments as part of the "Classification Experiments". And finally, we will use train a ResNet 50 to identify which earthworms are the same specimen. For this we will use the last layer of the neural network as embeddings that can be optimized according to the loss function "Triplet loss". We will refer to these experiments as "Identification Experiments".

2 Related work

Research specifically focused on earthworms remains limited. In particular, there is a notable lack of studies addressing earthworm classification involving more than three classes. In the context of computational taxonomic classification of earthworms using digital images, one of the most relevant studies is by Andleeb et al. [2021], who attempted to classify earthworms with machine learning techniques applied to digital images. According to the authors, this was the first attempt to classify earthworms from digital images. The study compared handcrafted image features, such as SIFT (Scale Invariant Feature Transform), Haar features, and HOG (Histogram of Oriented Gradients), with deep learning features. Both types of features were evaluated using machine learning classifiers, including Support Vector Machine (SVM) and Random Forests. The results indicated that deep learning features outperformed handcrafted features in earthworm classification. However, this work was limited to binary classification. Another study Malewski et al. [2025] also utilized handcrafted features for classifying earthworms from digital images. In this research, Malewski et al. [2025] extracted texture parameters from individual color channels (R, G, B, L, a, b, X, Y, Z, U, V, and S) using images of three genera: *Eisenia fetida*, *Dendrobaena* ssp., and *Lumbricus terrestris*. They reported 100 percent accuracy using machine learning methods such as Logistic Regression, Ensemble, and Narrow Neural Network. However, the study did not provide the digital images, and its reliance on color features may be problematic, as coloration can vary during different developmental stages of earthworms.

3 Methods

In this section, we will describe the datasets used in this project, then detail the data preprocessing steps, followed by the data augmentation techniques utilized. Next, we will explain the model evaluation metrics used to assess classification and identification performance, and finally, discuss the data balancing techniques applied to address imbalanced datasets.

3.1 Dataset

3.1.1 UK Dataset

This dataset contains approximately 11,827 images of earthworms collected in the UK from 2023 to 2024 by The Biological Recording Company and the UK Centre for Ecology & Hydrology, funded by Defra Brown [2024]. Images were taken following a specific protocol, where earthworms were placed in a clean white or black container, photographed with a smartphone to include the entire container, with multiple photos taken for each specimen. Experts later identified and annotated the classifications of the earthworms.

The original images, in .jpg format (see Figure 1 two left images), were reduced to 1,478 after curation and preprocessing. These images represent five genera: *Lumbricus*, *Aporrectodea*, *Alolobophora*, *Eisenia*, and *Eiseniella*. The class distribution is highly imbalanced, with *Aporrectodea* having over 20 times more images than *Eiseniella*. The dataset was split into approximately 65% training, 21% testing, and 15% validation (see Table 1), ensuring that images of the same individual appeared in only one set to prevent data leakage.

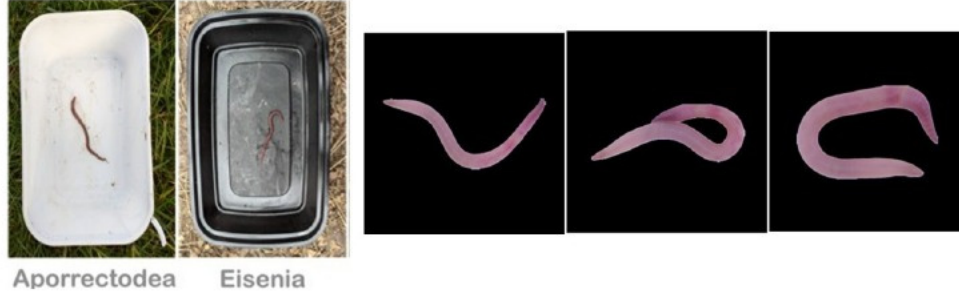


Figure 1: Two left images: Examples of the raw data used in the UK Dataset. The earthworms were imaged in a black or white container with smartphone cameras in different areas of the United Kingdoms. Three right images: Examples of the data used in the DK Dataset. The earthworms were segmented so the background was black. This figure shows an example of an earthworm photographed three times in different positions. This is the same specimen and in this project we will use this information about the same earthworm to distinguish it from other individual specimens, so that the identifies each individual as an individual.

Table 1: Number of individual earthworms of each genera used for the train, validation and test splits. "Val" stands for validation. As shown, the dataset is highly imbalance, containing 36 samples of the genus *Eiseniella*, and 758 samples of the genus *Aporrectodea*.

Genus	Train	Val	Test	Total
<i>Lumbricus</i>	300	60	96	456
<i>Aporrectodea</i>	503	102	153	758
<i>Allolobophora</i>	108	27	33	168
<i>Eisenia</i>	42	6	12	60
<i>Eiseniella</i>	20	6	10	36

3.1.2 DK Dataset

This dataset contains images of earthworms collected in Denmark, produced at Aarhus University in collaboration with the Department of Ecoscience and the Center for Quantitative Genetics and Genomics. The images were recently gathered and were not used in the initial phase of this project. It includes around 4,268 images (see Figure 1) from 775 individual earthworms, with an average of 6 images per worm. The data is divided into approximately 3,037 instances for training (71%), 633 for testing (15%), and 604 for validation (14%).

3.2 Data preprocessing

Many images within UK Dataset presented challenges due to low resolution and repeated content. To address these issues, we implemented an initial cleaning process. From the total of 11,827 images, which included numerous images of the same earthworms, we selected 1478 earthworm images belonging to the genera *Aporrectodea*, *Eisenia*, *Eiseniella*, *Lumbricus*, and *Allolobophora*. This method was intended to prevent redundancy within the dataset. Subsequently, we cropped the images to the size of the earthworm, ensuring the entirety of the earthworm's body remained visible. We then employed the deep learning algorithm known as "Segment Anything" (Kirillov et al. [2023]) to segment the earthworms from their backgrounds, ensuring that only the anatomical features of the earthworms were discernible to the deep learning models. This segmentation was vital in mitigating any influence from the background, such as distinctions between the white and black container or the presence of dirt in the containers, allowing the models to focus exclusively on the anatomical characteristics of the earthworms. Following the segmentation, we applied a cropping technique utilizing a bounding box to identify the smallest rectangular area that contained the earthworm. This approach reduced the proportion of the black background relative to the earthworm itself. Once the cropping process was concluded, we proceeded with data augmentation. A flowchart depicting the data preprocessing steps is included in the accompanying figure 21).

To study the impact of the background area over performance, we created two additional scenarios by rotating the cropped images at various angles and adjusting them to a square format. In the first scenario, we rotated the image to maximize the corresponding square background area (testing angles ranging from 0 to 90 degrees), while in

the second scenario, we rotated the image to minimize the corresponding square background area (also testing angles from 0 to 90 degrees) (see Figure 2 4)). The rationale for testing square areas was to avoid distortion when resizing the images to dimensions of 224 x 224 pixels, which is the designated input size for our models.

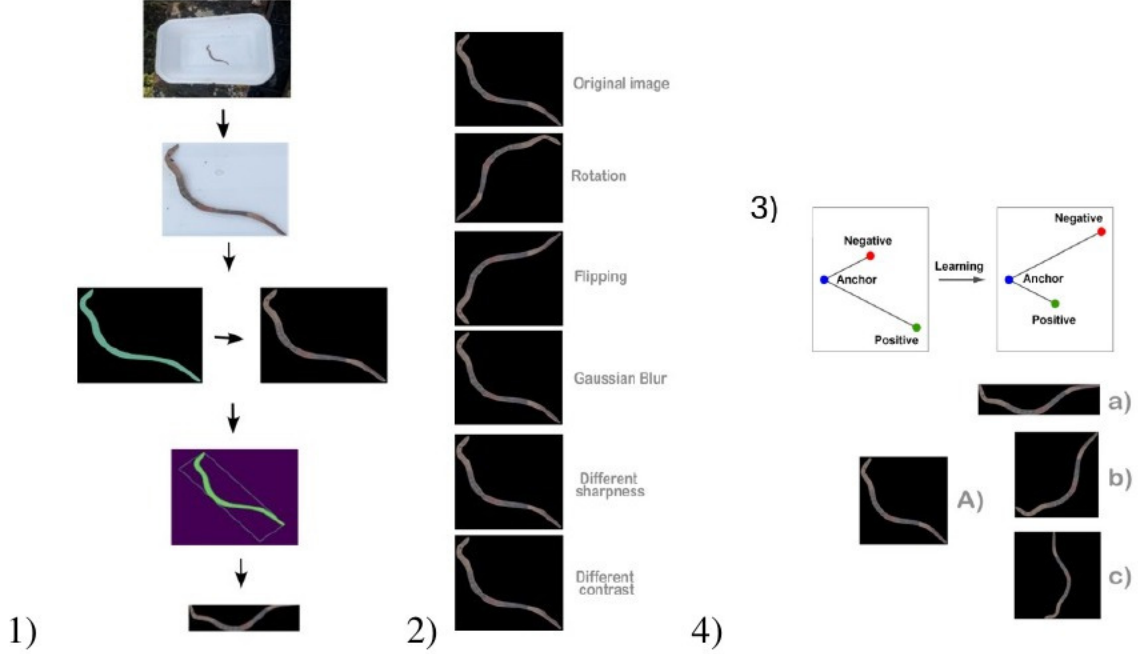


Figure 2: 1) Shows the data preprocessing workflow: The raw image is cut to the size of the earthworm. Afterwards, a segmentation mask is created with Segment Anything. This is used to segment the earthworm removing the background information. Thereafter, a bounding box detecting the minimum rectangle area around the earthworm is used to cut the earthworm to reduce the background area to the minimum. 2) Shows the different augmentation techniques used in this project: Rotation, flipping, gaussian blur, change in sharpness and in contrast. 3) Triplet loss illustration (explained in the section "Experiments", set here by lack of space). The goal is to train a model so that the distance between positive and anchor is smaller than from negative to anchor plus a margin, in the embedding space. 4) Shows the 3 cases in which we experiment with background area and how changes in the amount of pixels of the background influences performance. Given an original image of the size A) the 3 cases are: a) b) minimum rectangle area, minimum square area and c) maximum square area.

3.3 Data Augmentation

We employed several simple augmentation techniques to simulate realistic images of earthworms, ensuring that the model encountered images representative of the original data distribution. The augmentation methods used included rotation, flipping, Gaussian blur, adjustments in sharpness, and changes in image contrast (see Figure 2 2)). This approach allowed us to expand our dataset, which was limited in size, as deep learning models require large amounts of data for effective training. Additionally, data augmentation served as a strategy to reduce overfitting, as it enabled the model to learn from a more diverse set of images, ultimately improving generalization.

3.4 Model Evaluation

The model was evaluated using the F1-score as the primary metric, with accuracy also considered as an overall metric. The F1-score measures how well each class is classified, making it particularly useful for unbalanced datasets like ours. It accounts for the classification of smaller classes, ensuring that a high score does not solely stem from accurately classifying a more abundant class.

On the other hand, accuracy measures the overall performance by indicating the proportion of correctly classified images, regardless of class distribution. While accuracy can be informative, it has a significant drawback: if a model performs well on the classes with more instances, it may achieve a high accuracy score even if it fails to classify the smaller classes effectively.

Formally, accuracy can be defined as follows:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

The F1-score is the harmonic mean of precision and recall. Precision indicates how many of the predicted positive instances are actually correct, while recall measures how many actual positive instances were correctly identified, taking into account both those predicted as positive and those predicted as negative that are actually positive.

$$\text{Recall} = \frac{TP}{TP + FN} \quad \text{Precision} = \frac{TP}{TP + FP} \quad \text{F1Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

For the identification experiments we evaluated the model solely on the accuracy measurements.

3.5 Data balancing

Given that our dataset is imbalanced, we aim to equalize the exposure of the minority classes to the model by oversampling these classes in the training batches. This approach ensures that the model encounters the minority classes more frequently. We utilize the *WeightedRandomSampler*, a PyTorch implementation designed to assign varying weights to each sample in the training dataset. This way, samples from the minority classes are more likely to be included in the batches compared to those from the majority classes. The weights are determined by assigning to each sample from a particular class the inverse of the total number of samples in that class: $\text{weight}_{class} = \frac{1}{\text{Total number of samples in that class}}$.

4 Experiments

4.1 Classification experiments

We compared three neural networks for classification: ResNet-50, Swin Transformer, and ConvNeXt. ResNet-50 was chosen for its medium size, making it suitable for our small dataset. Similarly, we selected the basic architecture of the Swin Transformer. Our objectives were to evaluate the performance of these models in classifying earthworms and to assess the effects of background area and distortion on classification accuracy.

ResNet-50 served as the baseline model, while ConvNeXt was selected for its convolutional design that mimics Swin Transformers. We hypothesized that ConvNeXt might outperform ResNet-50. The Swin Transformer was included as a state-of-the-art model. All three models were pretrained on the ImageNet 1K dataset and then fine-tuned on our specific dataset to enhance their performance on the classification task.

In classifying earthworm images into categories like *Eisenia*, *Eiseniella*, *Alollobophora*, *Aporrectodea*, and *Lumbricus*, we aimed to reduce background area to improve accuracy. We believed that minimizing the background would help the models focus more on the earthworm itself. Fixed image dimensions (224 x 224 pixels) could lead to resolution loss, affecting classification due to background size.

We varied hyperparameters such as learning rate, weight decay, and dropout, achieving optimal results with a learning rate of 0.00001, a weight decay of 0.005, and a dropout rate of 0.5. The Adam Optimizer outperformed Stochastic Gradient Descent. We used cross-entropy loss and modified the fully connected layers of each model to classify five classes, adjusting ResNet-50's layer from (2048, 1000) to (2048, 5). Finally, we implemented early stopping a patience of 5 and a delta of 0.001.

4.2 Identification experiments

The objective of the identification experiments was to evaluate how effectively a deep learning model can determine whether two images feature the same earthworm. We utilized ResNet-50 for this purpose, training the model such that its last fully connected (fc) layer outputs a 128-dimensional embedding. This embedding represents the location in the dimensional space where, after training, we expect that the images of the same individual are grouped.

To train this model, we used three images: an anchor image, a positive image, and a negative image. The anchor image is the earthworm image we want to assess; the positive image contains another picture of the same earthworm, and the negative image features a different earthworm. The goal is for the distance in the embedding space between the anchor and the positive image to be less than the distance between the anchor and the negative image, ensuring that images of the same earthworm specimen are closer together than images of different specimens.

To accomplish this, we optimized the model using the "Triplet Loss" function, which measures the distances between the anchor and the positive image, and between the anchor and the negative image. After training, the positive embedding

should be closer to the anchor than the negative embedding plus a margin (typically set to 0.2) (see Figure 23). The distance between the anchor and positive embeddings is computed using the L2 norm in Euclidean space, while the distance between the anchor and negative embeddings is determined in a similar manner. The loss function is then defined in relation to these distances, with a margin denoted by alpha. If the distance between the anchor and the positive embeddings, which are defined as $f(A)$ and $f(P)$ respectively, is measured by the L2 norm in the euclidean space, we can define this distance as $\|f(A^{(i)} - f(P)^{(i)}\|_2^2$. And the distance between anchor and negative embeddings, $f(A)$ and $f(N)$ respectively, defined as $\|f(A^{(i)} - f(N)^{(i)}\|_2^2$. Then, the loss function is determined as

$$L = \sum_{i=1}^m \max(\|f(A^{(i)} - f(P)^{(i)}\|_2^2 - \|f(A^{(i)} - f(N)^{(i)}\|_2^2 + \alpha, 0)$$

where α is the margin.

In these experiments, the Adam optimizer was used for its better performance, with a learning rate of 0.0001 and an exponential scheduler set to gamma = 0.8. The ResNet-50 architecture was modified to adjust the fully connected layer from (2048, 1000) to (2048, 128) features, where 128 is the embedding space. The triplet loss function was applied, and t-SNE was utilized for dimensionality reduction to visualize the embedding distances in 2D, as shown in the Figure 4B).

5 Results

5.1 Classification results

The experimental results, which compare three models (ResNet 50, ConvNeXt, and Swin Transformer), are summarized in Table 2. Various configurations were evaluated, including images with minimal background rectangular areas, images rotated to achieve minimal background square areas, and images designed to maximize background square areas. On average, training and validation accuracies for most models ranged between 70% and 80%, which is notable given the lower image resolution. The minimal difference between training and validation accuracies and losses indicates that overfitting is not present. According to Table 2, the Swin Transformer achieved the highest performance in genus classification, with an F1 score of 0.700 and an accuracy of 75.9% in experiments using minimum square background area images. ConvNeXt ranked second, attaining an F1 score of 0.647 and an accuracy of 74% in the same experimental setting. These results demonstrate that the Swin Transformer is an effective classification network, excelling in both larger and smaller classes, as evidenced by its highest accuracy and F1-score in the test set. Additionally, experiments varying the background area and image distortion indicate that minimizing background area without distorting the image yields the best classification results. This outcome is logical, as a larger background area during resizing reduces image resolution, and image distortion complicates the identification of similarities among earthworm images belonging to the same genus. Furthermore, background area appears to have a greater impact on classification performance than distortion, a trend consistent across all three models. This is supported by the observation that test results for the minimum rectangle area configuration are higher than those for the maximum square background area. Although the minimum rectangle area introduces some distortion, its smaller background area suggests that loss of resolution or increased background area reduces performance more than moderate distortion. The accompanying figure 3 presents accuracy and loss plots, illustrating that loss decreases and accuracy increases in both training and validation sets of, for example, ResNet 50, confirming that the model is learning without overfitting. The confusion matrix for the Swin Transformer using minimum square background area images further demonstrates strong performance, particularly in predicting the largest class (Figure 3).

5.2 Identification results

The ResNet-50 model was trained for 10 epochs. As shown in Figure 4A), both the loss and accuracy improved during training, indicating that the model was learning effectively. However, signs of overfitting emerged around epoch 7, where the training loss continued to decrease while the validation loss plateaued. This divergence is also evident in the accuracy plot, where validation accuracy stabilized while training accuracy continued to improve slightly. Notably, the validation loss remained below the training loss throughout most of the training process. This phenomenon may be attributed to the substantially larger size of the training set compared to the validation set, potentially resulting in greater intra-class variability when computing embedding distances in the training data. This suggests a distributional difference between the training and test sets. Nevertheless, the primary objective—that both validation and training losses decrease—was achieved.

The model demonstrated strong performance in differentiating individual earthworms by measuring distances in the embedding space. Initial validation accuracy was approximately 63%, indicating that even before training, the

Table 2: Classification results using the ResNet 50 , ConvNeXT and Swin Transformer models in experiments with the minimum background rectangle area, minimum square background area and maximum square background area.

Model	Experiment	Train accuracy	Validation accuracy	Test accuracy	Test F1 score
ResNet 50	- Min. rectangle	0.763	0.706	0.651	0.586
	- Min. square	0.781	0.731	0.720	0.633
	- Max. square	0.797	0.756	0.668	0.536
ConvNeXT	- Min. rectangle	0.782	0.711	0.701	0.623
	- Min. square	0.775	0.741	0.740	0.647
	- Max. square	0.748	0.705	0.694	0.516
Swin T	- Min. rectangle	0.765	0.717	0.717	0.620
	- Min. square	0.789	0.760	0.759	0.700
	- Max. square	0.714	0.711	0.710	0.580

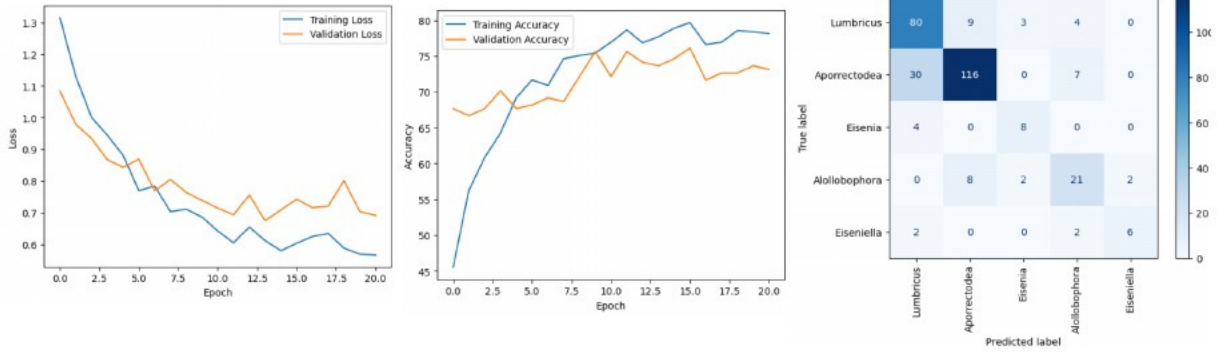


Figure 3: **Left:** Loss plot of the experiments performed with ResNet 50 using images with minimum square background area. As shown the validation and the training loss decrease until epoch 20 which was stopped by early stopping. **Center:** Accuracy plot of the experiments performed with ResNet 50 using images with minimum square background area. As show the accuracy increases in the validation and training data stabilizing in between 70 and 80 %. **Right:** Confusion matrix for the Swin Transformer from the minimum square background area experiments. As shown , the model classifies well the most abundant classes and fails to classify well the smaller classes, however it performs better than the other models.

embeddings of anchor images were closer to positive images (other images of the same earthworm) than to negative images (images of different earthworms). After training, performance improved substantially, reaching 92% accuracy on the training set and 91% accuracy on the validation set.

Further analysis using t-SNE dimensionality reduction to visualize embeddings in two-dimensional space (Figure 4B)) corroborates these findings. Examining 10 sample earthworm individuals from the validation set (left panel), with two images per individual, reveals that the model consistently predicts images of the same individual as proximate in the embedding space—points corresponding to the same individual (represented by the same color) cluster closer to each other than to other points. The same pattern was observed in the test set (right panel), where 10 different individuals were analyzed, demonstrating that embeddings of the same individual cluster together in the embedding space, even though in the test set this pattern was reduced.

6 Discussion

In this project, we successfully classified earthworms into their respective genera with an accuracy of 75.9% and an F1 score of 0.7. Our investigation into the effects of background area and image distortion revealed that the optimal performance was achieved with minimal background area and undistorted square images, whereas maximal square areas yielded significantly poorer results. Furthermore, we found that utilizing triplet loss during training allowed for an impressive estimation of around 90% accuracy in determining whether different images belonged to the same individual earthworm.

The Swin Transformer model outperformed other architectures for genus classification, yet it still achieved only 75.9% accuracy and an F1-score of 0.7. This could be attributed to the dataset’s substantial class imbalance, notably affecting the smaller genera (Eisenia and Eiseniella). To address this issue, it would be prudent to explore advanced strategies

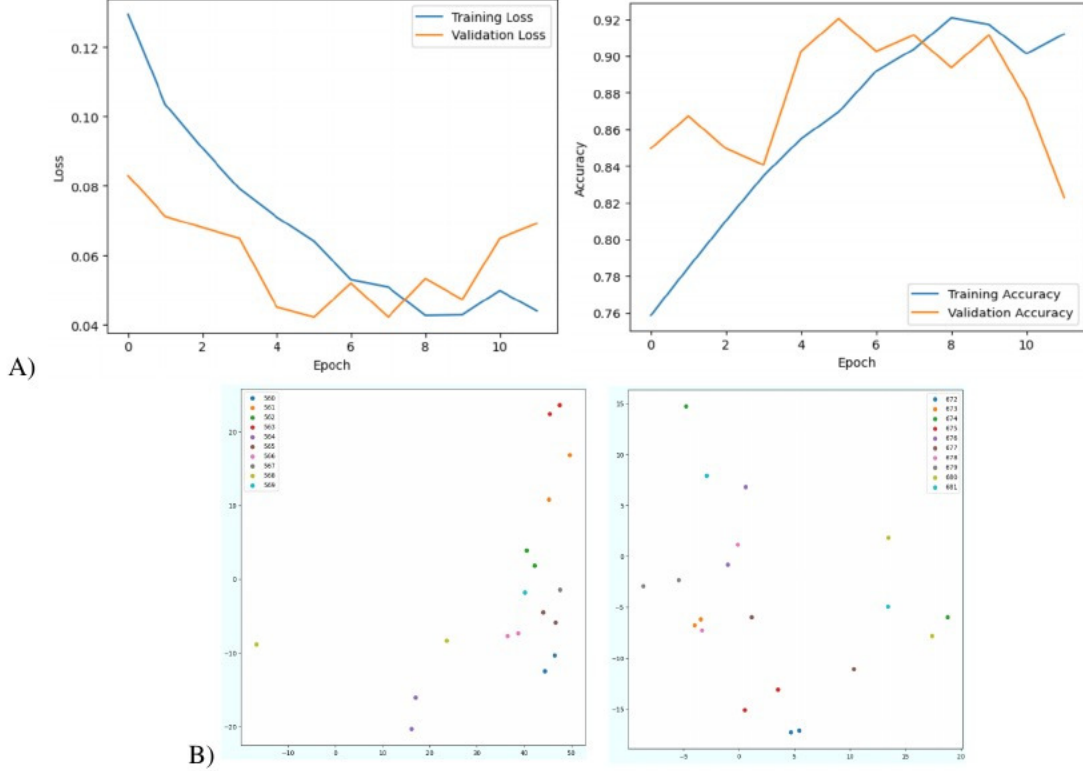


Figure 4: A) Loss and accuracy plots for the identification experiments trained on ResNet50 for 10 epochs. As shown both the training and validation accuracies increase and the losses decrease. B) Visualization of the embeddings after tSNE reduction to 2D of the validation set (left) and the test set (right). As shown many of the points belonging to the same individuals are close to each other in the embedding space, suggesting that the model has learned to classify which images belong to the same individual earthworm, especially in the case of the validation set.

for managing imbalanced datasets rather than merely increasing the frequency of smaller classes in training batches. One promising avenue is to implement the Focal Loss function, which is an improved version of cross entropy loss for imbalanced datasets.

Additionally, the classification dataset is relatively limited, comprising only 1,478 images, some of which feature the same earthworm specimen. This presents constraints for the model’s learning, particularly concerning the smaller classes with merely 36 and 60 images available, despite attempts to augment the dataset. To enhance dataset size and variety, we propose incorporating the newly collected DK dataset alongside the existing UK dataset and supplementary images from the Natural History Museum in the UK. This expansion would provide a more robust foundation for the models to learn and generalize the classification patterns.

Moreover, exploring the potential of synthetic image generation for data augmentation could further amplify our dataset. It is noteworthy that we achieved a 75.9% accuracy and 0.7 F1-score based solely on 1,478 images. Enhancing classification results may also involve utilizing higher-resolution images or focusing on specific detailed segments of earthworm anatomy, as features that are often overlooked by smartphone cameras could prove crucial for accurate classification. The concept of providing a sequence of images to the model could facilitate the association of detailed anatomical features with specific classes.

For identification tasks, an additional strategy to improve results would be to evaluate alternative models and to conduct further assessments on whether multiple images of the same specimens are clustered closely in the embedding space (so far, we have tested only two). Exploring the effects of introducing new specimens could also yield insights into whether the new embeddings sufficiently diverge from those of the trained embeddings, ensuring that clustering algorithms do not erroneously group dissimilar specimens together.

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